

Project Proposal: Automatic MRI View Classification and Cancer Type Prediction Using Unsupervised Clustering and Ensemble Learning

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Introduction

Medical imaging plays an important role in the early detection and diagnosis of cancer. Magnetic Resonance Imaging (MRI) provides detailed pictures of internal body structures, but MRIs are taken from different angles.

These different views can make it difficult for a single machine learning model to correctly identify patterns of cancer. If we first understand which view an MRI belongs to, we can then use a more suitable model for that specific view. This can make cancer detection more accurate.

In this project, we aim to build a system that can automatically group MRI images according to their view type using unsupervised learning, and then use ensemble learning to predict the type of cancer.

Problem Being Addressed

MRI images come from multiple orientations, and this variety confuses normal classification models. A single model trained on mixed orientations performs poorly.

We need a method to first separate images based on their view type (orientation) before predicting the cancer type. Our goal is to automate this process without manual labeling of views.

Objectives

The main objectives of the project are:

1. To automatically cluster MRI images into groups based on their orientation (front, side, top) using K-Means clustering.
2. To extract simple shape-based features that describe each MRI image.
3. To build an ensemble model that predicts the type of cancer after the orientation is known.
4. To create a complete pipeline that can take any new MRI image and automatically classify it into its orientation and then predict the cancer type.

Methodology

Our project will have two stages:

Stage 1: Unsupervised Clustering of MRI Views

- Convert MRI images to grayscale and remove the background.
- Extract shape-based features such as:
 - Aspect Ratio: width/height of the head.
 - Fill Ratio: how much of the image is covered by the head.
 - Left-Right Symmetry: how similar the left and right halves are.
 - Top-Bottom Symmetry: similarity between top and bottom halves.
- Combine these values into a small feature vector for each image.
- Use K-Means clustering ($k=3$) to automatically group the images into three view types:
 - Cluster 0 → Front or back view
 - Cluster 1 → Side view
 - Cluster 2 → Top view

Stage 2: Ensemble Learning for Cancer Type Prediction

- After the orientation of an image is known, it is passed to the corresponding model trained for that view.
- Each model will be an ensemble (a combination of different classifiers).
- The ensemble will predict the type of cancer with improved accuracy.

Expected Results

We expect the unsupervised clustering step to group MRI images correctly according to their view. Then, the ensemble learning models will give more accurate cancer predictions because each model will specialize in one type of MRI view.

Overall, the system will make the cancer detection process more automatic and reliable.

Tools and Technologies

- Programming Language: Python
- Libraries: OpenCV, NumPy, Scikit-learn, TensorFlow/PyTorch

- Algorithms: K-Means Clustering, Ensemble Learning

Dataset

<https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset?resource=download>